



Versatile value-added application of hyperbranched lignin derivatives: Water-resistance adhesive, UV protection coating, self-healing and skin-adhesive sensing

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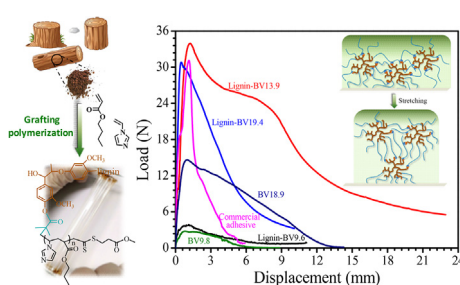
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HIGHLIGHTS

- The versatile lignin copolymers were synthesized via a surface RAFT polymerization.
- Hyperbranched structure of lignin provides improved adhesive performances.
- Adhesive performances could be well controlled by altering the content of VI.
- Room temperature self-healing elastomer was achieved by appropriate VI content.

GRAPHICAL ABSTRACT



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ABSTRACT

Bio-based copolymers with versatile properties and good processability are highly desired for practical applications. Herein we describe novel hyperbranched adhesive, UV protective coatings and functional elastomers with 6.1–9.5 wt% lignin synthesized by a one-pot RAFT grafting polymerization. The unique macromolecular structure and sufficient hydrogen bonds of these Lignin-graft-poly(*n*-butyl acrylate-co-1-vinylimidazole) copolymers (Lignin-BVs) resulted in fascinating and controllable features. Specifically, incorporating 13.9 wt% imidazole (VI) effectively improved the adhesive performances with maximal adhesive force and strength up to 34.0 N and 367.8 N·s, respectively, which were better than the commercial 3 M double-sided adhesive. Owing to water-resistance and UV absorption capacity, our Lignin-BVs could be served as excellent adhesive coatings for future outdoor decorative materials, safety clothing or beach umbrella. Moreover, upon increasing VI content to 19.4 wt%, Lignin-BV19.4 elastomer showed automatically room temperature self-healing behavior and could be fabricated into skin-adhesive strain sensor. Our studies provided a green and effective method in the preparation of the high value-added lignin copolymers, which could be appealing to both frontier research and industrial development.

1. Introduction

With the implementation of the strategy for green and sustainable

development, manufacturing high-performance biomass materials with both versatile properties and commercialization applications has been a persistent target for recent decades [1–3]. Being the abundant

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